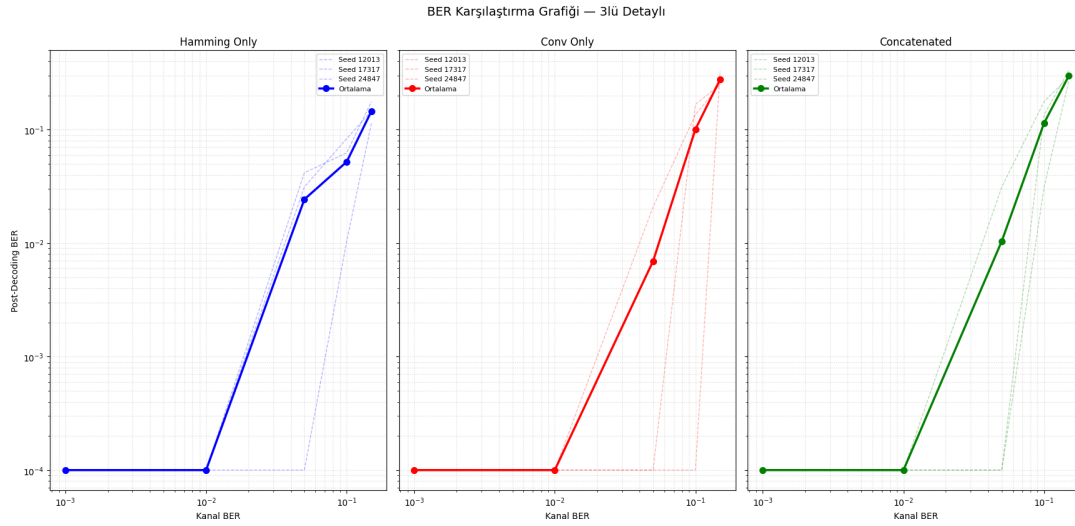
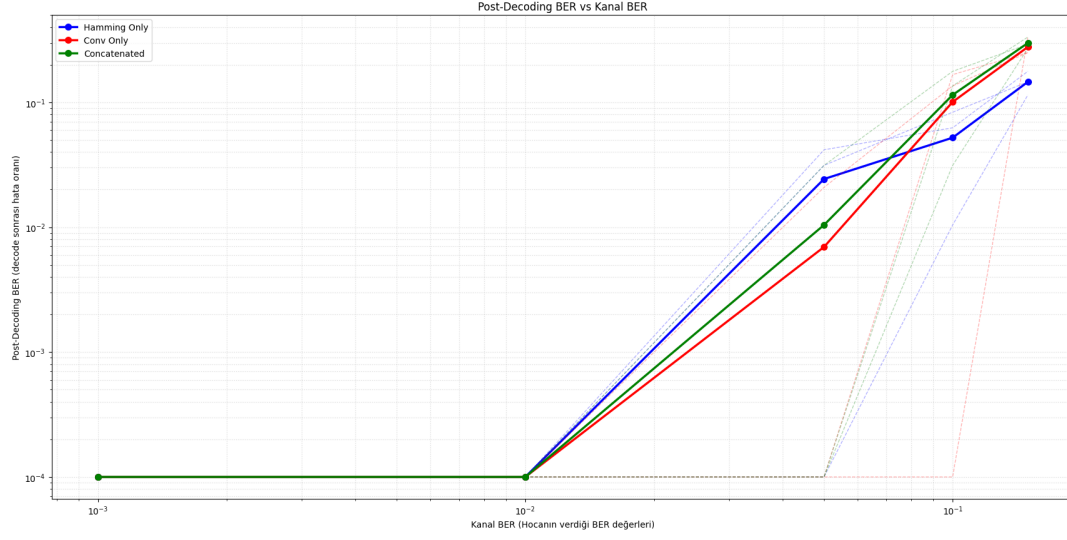


EE431 Phase3

Analysis

I chose to plot all the BER graphs in my project on a logarithmic scale. The main reason for this is that the BER values after post-decoding range from 10^{-4} to 10^{-1} , a very wide range. This graph, which also conforms to communication engineering standards, is available in both single and detailed versions on my GitHub repository.



Question 1

At which channel BER does the concatenated scheme begin to outperform each single scheme? Why might this crossover happen where it does?

In the graph, moving right means more noise, moving up means more errors.

```
PS C:\Users\burak\ee431_project> python -u "c:\Users\burak\ee431_project\analiz.py"

Post-Decoding BER ortalamalari:
Yontem          0.001    0.01    0.05    0.1    0.15
Hamming Only    0.00010  0.00010  0.02431  0.05208  0.14583
Conv Only       0.00010  0.00010  0.00694  0.10069  0.27778
Concatenated    0.00010  0.00010  0.01042  0.11458  0.29861
PS C:\Users\burak\ee431_project> █
```

Hamming Only

Our message is ECC2026-S09I, 96 bits. Since 96 is not divisible by 11, we appended 3 zero bits (padding), making it 99 bits. The Hamming code has $t = 1$, meaning it can correct at most 1 bit error per block. We divided 99 bits into 9 blocks of 11 bits each. Each block was Hamming encoded to 15 bits. There are 9 blocks of 15 bits passing through the channel, totalling 135 bits. The BSC channel corrupts each bit independently. BER= 0.05 means each bit has a 5% probability of being corrupted.

- BER= 0.001 $\rightarrow 15 \times 0.001 = 0.015 \rightarrow$ almost no errors per block $\rightarrow$ Hamming corrects easily
- BER= 0.01 $\rightarrow 15 \times 0.01 = 0.15 \rightarrow$ still very few errors $\rightarrow$ Hamming corrects
- BER= 0.05 $\rightarrow 15 \times 0.05 = 0.75 \rightarrow$ average 0.75 errors per block $\rightarrow$ some blocks receive 2 errors $\rightarrow$ Hamming can only correct 1 bit, becomes helpless when 2 errors arrive
- BER= 0.10 $\rightarrow 15 \times 0.10 = 1.50 \rightarrow$ average 1.5 errors per block $\rightarrow$ most blocks have 2+ errors $\rightarrow$ Hamming cannot correct the majority
- BER= 0.15 $\rightarrow 15 \times 0.15 = 2.25 \rightarrow$ almost every block has more than 2 errors $\rightarrow$ Hamming is ineffective

Since Hamming can only recover 1 error, when BER reaches 0.05, blocks start receiving 2 errors and the system becomes insufficient. This is why the blue curve in the graph rises at BER= 0.05.

Conv Only

Our message is 96 bits. The Conv encoder added 3 termination bits, making it 99 bits. Rate $1/2 \rightarrow 99 \times 2 = 198$ bits passed through the channel. Viterbi decoded the 198 bits, the last 3 termination bits were discarded, returning to 96 bits. The BSC channel corrupts each bit independently. BER= 0.05 means each bit has a 5% probability of being corrupted.

- BER= 0.001 $\rightarrow 198 \times 0.001 = 0.198 \rightarrow$ almost no errors $\rightarrow$ Viterbi comfortable
- BER= 0.01 $\rightarrow 198 \times 0.01 = 1.98 \rightarrow$ few errors, falling far apart $\rightarrow$ Viterbi corrects
- BER= 0.05 $\rightarrow 198 \times 0.05 = 9.9 \rightarrow \sim 10$ errors but spread across 198 bits $\rightarrow$ errors are far apart $\rightarrow$ Viterbi recovers most $\rightarrow$ 2/3 seeds successful (seed= 24847 failed, 2 errors remained)
- BER= 0.10 $\rightarrow 198 \times 0.10 = 19.8 \rightarrow 20$ errors, errors starting to fall close together $\rightarrow$ Viterbi selects wrong path $\rightarrow$ Seed= 12013: 16 errors remained, Seed= 24847: 13 errors remained, only Seed= 17317 succeeded
- BER= 0.15 $\rightarrow 198 \times 0.15 = 29.7 \rightarrow 30$ errors, errors side by side $\rightarrow$ Viterbi completely insufficient $\rightarrow$ all 3 seeds: Perfect=False

While Hamming fails at BER= 0.05, Conv holds until BER= 0.10. The reason is simple: as long as errors are spread out, Viterbi recovers them; when they cluster together, it fails.

Concatenated

Our message is 96 bits. First, Hamming encoding was applied, producing 135 bits. Then Conv encoding was applied, 3 termination bits were added $\rightarrow$ 138 bits. Rate $1/2 \rightarrow 138 \times 2 = 276$ bits passed through the channel. During decoding, first Viterbi (276 $\rightarrow$ 138 bits), then Hamming decode (135 $\rightarrow$ 96 bits) was applied. The BSC channel corrupts each bit independently. BER= 0.05 means each bit has a 5% probability of being corrupted.

- BER= 0.001 $\rightarrow 276 \times 0.001 = 0.276 \rightarrow$ almost no errors $\rightarrow$ Viterbi cleans, Hamming also cleans $\rightarrow$ perfect
- BER= 0.01 $\rightarrow 276 \times 0.01 = 2.76 \rightarrow$ few errors $\rightarrow$ Viterbi recovers, Hamming cleans remaining errors $\rightarrow$ perfect
- BER= 0.05 $\rightarrow 276 \times 0.05 = 13.8 \rightarrow \sim 14$ errors $\rightarrow$ Viterbi recovers most $\rightarrow$ Hamming cleans the rest $\rightarrow$ 2/3 seeds successful (seed= 24847 failed, 3 errors remained)
- BER= 0.10 $\rightarrow 276 \times 0.10 = 27.6 \rightarrow \sim 28$ errors $\rightarrow$ Viterbi starts making wrong corrections $\rightarrow$ corrupted bits arrive at Hamming $\rightarrow$ Hamming also corrects wrongly $\rightarrow$ two layers worsen each other $\rightarrow$ all 3 seeds failed
- BER= 0.15 $\rightarrow 276 \times 0.15 = 41.4 \rightarrow \sim 41$ errors $\rightarrow$ both Viterbi and Hamming completely insufficient $\rightarrow$ all 3 seeds failed

At BER= 0.10, Concatenated (11.5% error) performs worse than Conv Only (10% error). This is because more errors arrive since 276 bits pass through the channel, and Conv's wrong corrections send corrupted bits to Hamming, which corrupts them further. The two layers negatively affect each other.

****Concatenated failed to outperform both schemes simultaneously at any BER value.**

The Concatenated scheme outperforms Hamming Only at channel BER= 0.05. However, Concatenated never outperforms Conv Only at any BER value tested. In principle, Conv and Hamming support each other, but as BER increases, the hybrid structure loses its advantage.

Question 2

How does the empirical performance of your convolutional code relate to the theoretical d_{free} you computed in Phase 1?

d_{free}

Path 1: $a \rightarrow b \rightarrow c \rightarrow e \rightarrow a$

$$11 + 01 + 11 + 11 = 2 + 1 + 2 + 2 = 7$$

Path 2: $a \rightarrow b \rightarrow d \rightarrow g \rightarrow e \rightarrow a$

$$11 + 10 + 10 + 00 + 11 = 2 + 1 + 1 + 0 + 2 = 6^{***}$$

Path 3: $a \rightarrow b \rightarrow d \rightarrow h \rightarrow g \rightarrow e \rightarrow a$

$$11 + 10 + 01 + 01 + 00 + 11 = 2 + 1 + 1 + 1 + 0 + 2 = 7$$

$$d_{\text{free}} = 6$$

$$t = \left\lfloor \frac{d_{\text{free}} - 1}{2} \right\rfloor = \left\lfloor \frac{5}{2} \right\rfloor = 2$$

Viterbi can theoretically correct up to 2 bit errors.

- BER= 0.001 $\rightarrow 198 \times 0.001 = 0.198 \rightarrow$ almost no errors $\rightarrow t = 2$ easily sufficient
- BER= 0.01 $\rightarrow 198 \times 0.01 = 1.98 \rightarrow$ few errors, arriving spread out $\rightarrow t = 2$ sufficient
- BER= 0.05 $\rightarrow 198 \times 0.05 = 9.9 \rightarrow \sim 10$ errors $\rightarrow t = 2$ is strained but recovers most $\rightarrow 2/3$ seeds successful
- BER= 0.10 $\rightarrow 198 \times 0.10 = 19.8 \rightarrow \sim 20$ errors $\rightarrow t = 2$ exceeded $\rightarrow$ Viterbi selects wrong path $\rightarrow 1/3$ seeds successful
- BER= 0.15 $\rightarrow 198 \times 0.15 = 29.7 \rightarrow \sim 30$ errors $\rightarrow t = 2$ completely insufficient $\rightarrow 0/3$

$d_{\text{free}} = 6$ theoretically provides $t = 2$ capacity. Since Hamming Only has $t = 1$, it fails at BER= 0.05. Since Conv Only has $t = 2$, it holds until BER= 0.10. In other words, the larger the d_{free} value, the longer the system survives.

Question 3

Identify one BER value at which your concatenated scheme fails on at least one seed. Inspect a failing case in detail: where in the trellis did the Viterbi decoder go astray? Reproduce that section of the trellis by hand, annotated with path metrics.

Concatenated	0.001	12013	0	0	True	
Concatenated	0.001	17317	0	0	True	
Concatenated	0.001	24847	0	0	True	
Concatenated	0.010	12013	1	0	True	
Concatenated	0.010	17317	2	0	True	
Concatenated	0.010	24847	2	0	True	
Concatenated	0.050	12013	13	0	True	
Concatenated	0.050	17317	12	0	True	
Concatenated	0.050	24847	18	3	False	}
Concatenated	0.100	12013	29	13	False	
Concatenated	0.100	17317	21	3	False	
Concatenated	0.100	24847	32	17	False	
Concatenated	0.150	12013	45	32	False	
Concatenated	0.150	17317	30	26	False	
Concatenated	0.150	24847	51	28	False	

Concatenated | BER = 0.10 | seed = 24847 | 32 errors | 17 errors remain | False

I selected this case because the question asks for a Concatenated failure. The system clearly fails at BER= 0.10, and with 17 out of 32 errors remaining, it is ideal for observing where Viterbi went wrong. Additionally, I should mention that I did not fully understand the trellis for question 3 I only specified the criteria I need to base my trellis drawing on.